

解方程

$$x^3y - 4ydx + x^4dy = 0$$

配凑得到

$$4x^3ydx + x^4dy = 3x^3y + 4ydx$$

令 $u = x^4y$

则

$$du = \frac{3u}{x} + \frac{4u}{x^4}dx$$

$$\Rightarrow \frac{du}{u} = \left(\frac{3}{x} + \frac{4}{x^4} \right) dx$$

$$u = Cx^3e^{-\frac{4}{3x^3}}$$